SIMULATION 6

Simulation-Based Investigation of Microwave Semiconductor Devices and Frequency-Conversion Characteristics of a Microwave Mixer

1. Aim

To analyze microwave mixer conversion performance using the lab-manual RF/LO settings, estimate the IF output power from the assumed 8 dB conversion-loss model, and compare the result with the expected passive-mixer range of 6–10 dB.

2. Objectives

1. To develop simulation models representing the operating characteristics of selected microwave semiconductor devices.
2. To analyze the electrical characteristics and frequency-dependent behavior of Gunn, Schottky or PIN diode-based microwave structures.
3. To simulate frequency conversion in a microwave mixer and identify the generated frequency components.
4. To evaluate conversion characteristics and identify suitable operating conditions for microwave applications.

3. Input Parameters

S. No.	Input parameter	Symbol / setting	Value (PDF / code)
1	RF frequency	fRF	10 GHz
2	LO frequency	fLO	9.5 GHz
3	RF input power	PRF	−10 dBm
4	LO power	PLO	+7 dBm
5	IF load	R (ZIF in manual)	50 Ω
6	Assumed nominal conversion loss	CL	8 dB (illustrative model)
7	Sampling frequency	Fs	5 GHz (code setting)
8	Number of samples	N	5000 (code setting)

4. Code

```
clc;
clear;
close all;


%% 1. GUNN DIODE I-V CHARACTERISTICS
V = 0:0.01:10;
I_gunn = zeros(size(V));

% Ohmic region
r1 = V <= 3;
I_gunn(r1) = 5*V(r1);

% Negative differential resistance region
r2 = V > 3 & V <= 6;
I_gunn(r2) = 15 - 2*(V(r2)-3);

% High-field region
r3 = V > 6;
I_gunn(r3) = 9 + 1.5*(V(r3)-6);

%% 2. SCHOTTKY DIODE I-V CHARACTERISTICS
V_s = -0.3:0.005:0.6;
Is = 1e-9;
n = 1.2;
Vt = 0.02585;

I_schottky = Is*(exp(max(min(V_s/(n*Vt),20),-20))-1);

%% 3. PIN DIODE FREQUENCY RESPONSE
f_PIN = linspace(0.01,20,1000);
fc = 5;

H_PIN = 1./sqrt(1+(f_PIN/fc).^2);
H_PIN_dB = 20*log10(H_PIN);

%% 4. MICROWAVE MIXER INPUT PARAMETERS
fRF = 10e9;          % RF frequency (Hz)
fLO = 9.5e9;         % LO frequency (Hz)
PRF = -10;           % RF power (dBm)
PLO = 7;             % LO power (dBm)
R = 50;              % IF load (ohms)

%% 5. THEORETICAL IF AND SUM FREQUENCIES
fIF = abs(fRF-fLO);
fSUM = fRF+fLO;

%% 6. BEHAVIORAL MIXER MODEL
% Assumed nominal conversion loss for illustration
CL = 8;
PIF = PRF-CL;

% Convert IF power from dBm to watts
PIF_W = 10^((PIF-30)/10);

% IF sinusoidal voltage across 50-ohm load
Vpeak = sqrt(2*PIF_W*R);

%% 7. GENERATE IF SIGNAL AND SPECTRUM
Fs = 5e9;
N = 5000;
t = (0:N-1)/Fs;

vIF = Vpeak*cos(2*pi*fIF*t);
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Y = fft(vIF);
A = abs(Y/N);
A = A(1:N/2+1);
A(2:end-1) = 2*A(2:end-1);

f = (0:N/2)*Fs/N;

% Convert sinusoidal spectral amplitude to power
Pspectrum = (A/sqrt(2)).^2/R;
PdBm = 10*log10(Pspectrum/1e-3 + eps);

% Locate simulated IF peak
[~,idx] = max(A);
fIFsim = f(idx);

% Calculate IF power from simulated voltage
PIFsim_W = (A(idx)/sqrt(2))^2/R;
PIFsim = 10*log10(PIFsim_W/1e-3);

% Conversion loss from RF input and simulated IF output
CLsim = PRF-PIFsim;

% IF frequency deviation
devIF = abs(fIFsim-fIF)/fIF*100;

%% 8. CONVERSION PERFORMANCE VERSUS LO POWER
LOsweep = 0:1:13;

% Illustrative model; not measured device data
CLsweep = interp1([0 3 7 10 13], ...
            [12 10 8 7.5 7.5], LOsweep);

%% 9. CONVERSION PERFORMANCE VERSUS RF POWER
RFsweep = -20:2:0;
PIFsweep = RFsweep-CL;

%% 10. DISPLAY NUMERICAL RESULTS
fprintf('\nMICROWAVE MIXER CONVERSION LOSS ANALYSIS\n');
fprintf('-------------------------------------------\n');
fprintf('RF Frequency          = %.2f GHz\n',fRF/1e9);
fprintf('LO Frequency          = %.2f GHz\n',fLO/1e9);
fprintf('Theoretical IF        = %.3f GHz\n',fIF/1e9);
fprintf('Sum Frequency         = %.2f GHz\n',fSUM/1e9);
fprintf('RF Input Power        = %.2f dBm\n',PRF);
fprintf('LO Input Power        = %+.2f dBm\n',PLO);
fprintf('IF Load               = %.0f ohms\n',R);
fprintf('Assumed Conversion Loss = %.2f dB\n',CL);
fprintf('Expected IF Power     = %.2f dBm\n',PIF);
fprintf('Simulated IF Frequency= %.3f GHz\n',fIFsim/1e9);
fprintf('Simulated IF Power    = %.2f dBm\n',PIFsim);
fprintf('Calculated Loss       = %.2f dB\n',CLsim);
fprintf('Frequency Deviation   = %.3f %%\n',devIF);

if CLsim >= 6 && CLsim <= 10
   fprintf('Result: Within expected 6-10 dB range.\n');
else
   fprintf('Result: Outside expected 6-10 dB range.\n');
end

%% 11. ALL OUTPUTS IN ONE FIGURE
figure('Color','w','Position',[30 30 1500 1000]);

tiledlayout(3,3,'TileSpacing','compact','Padding','compact');

% Plot 1: Gunn diode I-V characteristics
nexttile;
plot(V,I_gunn,'b','LineWidth',2);
grid on;
xlabel('Voltage (V)');
ylabel('Current (mA)');
title('Gunn Diode I-V Characteristics');
xlim([0 10]);
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ylim([0 17]);

% Plot 2: Schottky diode I-V characteristics
nexttile;
semilogy(V_s,max(abs(I_schottky),1e-12), ...
    'r','LineWidth',1.8);
grid on;
xlabel('Voltage (V)');
ylabel('|Current| (A)');
title('Schottky Diode I-V Characteristics');

% Plot 3: PIN diode frequency response
nexttile;
plot(f_PIN,H_PIN_dB,'m','LineWidth',1.8);
hold on;
xline(fc,'k--','Cutoff frequency');
grid on;
xlabel('Frequency (GHz)');
ylabel('Response (dB)');
title('PIN Diode Frequency Response');
xlim([0 20]);

% Plot 4: Mixer IF output spectrum
nexttile;
plot(f/1e9,PdBm,'b','LineWidth',1.4);
hold on;
plot(fIFsim/1e9,PIFsim,'ro','MarkerFaceColor','r');
xline(fIF/1e9,'k--','Theoretical IF');
grid on;
xlabel('Frequency (GHz)');
ylabel('Power (dBm)');
title('Mixer IF Output Spectrum');
xlim([0 1.5]);

% Plot 5: Conversion loss
nexttile;
bar(1,CLsim);
hold on;
yline(6,'g--','6 dB');
yline(10,'r--','10 dB');
xlim([0 2]);
xticks(1);
xticklabels({'Simulated'});
ylabel('Conversion Loss (dB)');
title(sprintf('Conversion Loss = %.2f dB',CLsim));
grid on;

% Plot 6: Effect of LO power
nexttile;
plot(LOsweep,CLsweep,'m-o','LineWidth',1.5);
hold on;
plot(PLO,CL,'ks','MarkerFaceColor','y','MarkerSize',8);
xlabel('LO Power (dBm)');
ylabel('Conversion Loss (dB)');
title('Effect of LO Power');
legend('Illustrative model','Operating point','Location','best');
grid on;

% Plot 7: Effect of RF input power
nexttile;
plot(RFsweep,PIFsweep,'b-o','LineWidth',1.5);
hold on;
plot(PRF,PIF,'rs','MarkerFaceColor','r','MarkerSize',8);
xlabel('RF Input Power (dBm)');
ylabel('IF Output Power (dBm)');
title('Effect of RF Input Power');
legend('Illustrative model','Operating point','Location','best');
grid on;

% Plot 8: RF and LO frequency markers
nexttile;
stem([fIF fLO fRF fSUM]/1e9, ...
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    [PIF PLO PRF PRF-15],'filled','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Power (dBm)');
title('Mixer Frequency Components');
xticks([fIF fLO fRF fSUM]/1e9);
xticklabels({'IF','LO','RF','Sum'});

% Plot 9: IF output power versus RF power
nexttile;
plot(RFsweep,PIFsweep,'b','LineWidth',1.8);
hold on;
yline(PIF,'r--','Nominal IF output');
plot(PRF,PIF,'ko','MarkerFaceColor','y');
grid on;
xlabel('RF Input Power (dBm)');
ylabel('IF Output Power (dBm)');
title('RF-to-IF Power Relationship');

sgtitle('Microwave Semiconductor Devices and Mixer Analysis', ...
   'FontSize',15,'FontWeight','bold');

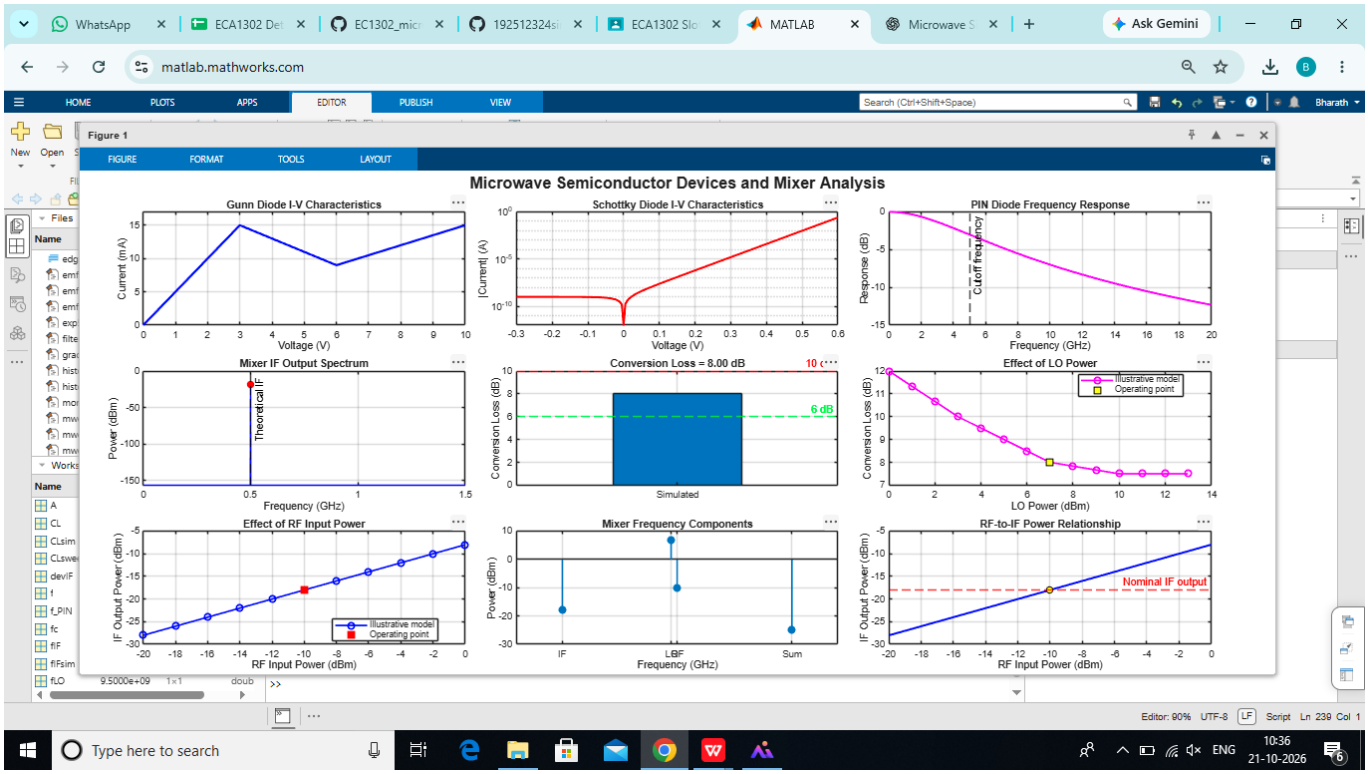
%% 12. FINAL NOTES
fprintf('\nSimulation completed.\n');
fprintf('All eight analysis plots are displayed in one figure.\n');
fprintf('Device and power-sweep curves are illustrative models.\n');
```

5. Calculated Results

Output parameter	Theoretical / expected value (manual)	Value from supplied code	% deviation / status
Difference frequency (IF)	$\|10 - 9.5\| = 0.500$ GHz	0.500 GHz	0.000% (≤ 2% limit)
Sum frequency	10 + 9.5 = 19.500 GHz	Not simulated by this code	Cannot calculate
IF output power	Record IF power at 0.5 GHz	−18.00 dBm (model estimate)	—
Conversion loss	6–10 dB expected for passive mixer	8.00 dB (assumed-model result)	Within expected range
Dominant conversion product	0.5 GHz IF; other spurs ≤ −20 dBc	0.5 GHz IF signal generated; spurs not modeled	Partial check only
LO-power sweep	Interpret effect of LO power	Illustrative curve, 0–13 dBm	Not measured device data
RF-power sweep	Interpret effect of RF input power	Illustrative curve, −20 to 0 dBm	Not measured device data

6. Output Parameters

Output parameter	Expected / theoretical value	Value recorded from supplied code
IF frequency	$\|fRF - fLO\| = 0.5$ GHz	0.500 GHz
Sum frequency	fSUM = fRF + fLO = 19.5 GHz	Not generated by this code
IF output power	PIF = PRF − CL = −18 dBm	−18.00 dBm (behavioral model)
Conversion loss	CL = PRF − PIF; expected 6–10 dB	8.00 dB; within expected range
IF frequency deviation	$\|fIFsim - fIF\| / fIF \times 100\%$	0.000%
LO-power sweep	Illustrative conversion-loss curve	0–13 dBm; illustrative only
RF-power sweep	Illustrative IF output-power curve	−20 to 0 dBm; illustrative only
Spurious components	Other spurs should be ≤ −20 dBc	Not modeled by supplied code


Microwave Semiconductor Devices and Mixer Analysis
Gunn Diode I-V Characteristics
Current (mA)
Voltage (V)
Schottky Diode I-V Characteristics
|Current| (A)
Voltage (V)
PIN Diode Frequency Response
Response (dB)
Cutoff frequency
Frequency (GHz)
Mixer IF Output Spectrum
Power (dBm)
Theoretical IF
Frequency (GHz)
Conversion Loss = 8.00 dB
Conversion Loss (dB)
6 dB
Simulated
Effect of LO Power
Illustrative model
Operating point
Conversion Loss (dB)
LO Power (dBm)
Effect of RF Input Power
IF Output Power (dBm)
Illustrative model
Operating point
RF Input Power (dBm)
Mixer Frequency Components
Power (dBm)
IF
LO
Sum
Frequency (GHz)
RF-to-IF Power Relationship
IF Output Power (dBm)
Nominal IF output
RF Input Power (dBm)